

DS 3010: Applied Data Modeling and Predictive Analysis (Fall 2025)

Project Proposal

1 Project Title

Your machine Learning project.

- RWIS-ML

2 Team Members

- Harrison Voss
- Alex Mears

3 Project Details

3.1 Project Objective

Please describe your project objective in this part.

- What is the project Objective?
 - The objective of the project is to train a model to identify weather conditions on Iowa roads.
- What problem to solve?
 - Given an image of a roadway and ambient weather information, classify the road condition, i.e; icy, snowy, wet, dry.
- Why the problem is important?
 - Driving on roads with adverse conditions is dangerous, so being able to identify the conditions accurately can help drivers be safer.
- Why machine learning can help to solve the problem?
 - The amount of data that is needed to solve the problem is too large for a human to program. Road conditions don't necessarily follow a nice pattern, so we need to use machine learning.

3.2 Datasets

Please describe your dataset in this section.

- What is the data and where you obtain it?
 - The data is archived weather data and webcam images from ISU Mesonet:
<https://mesonet.agron.iastate.edu/>
- How the data is collected?

- We will be scraping the realtime webcam images from the website. The historic weather data is downloadable from the website directly. The data was collected from the RWIS network of stations around Iowa.
- What will be the features and labels you will use?
 - We will use images of roadways taken from RWIS webcams, as well as the pavement and weather data from that same station. Air Temp, Pavement Temp, Pavement Condition.
- How many examples for training, validation, and testing?
 - Insert how many examples here. Once we have built the model, we will do an 80/20 training to initially validate and test it. Throughout the rest of the semester, we will continue to test and validate the model with the real time data.

3.3 Machine Learning Algorithm

Please describe the machine learning algorithm you want to use for your project.

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Please justify your selection.

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3.4 Expected Outcomes

What is your expected outcome for this project?

- To have a model that will be able to identify the condition of a road based on an image from an RWIS webcam with an accuracy of at least 80%